CSA02 – ASSIGNMENT 1

Grama Jyothi Rural Electricity Co-operative

TASK 1 – Understand the Problem and Identify the Data (4 Marks)

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1.1 Problem Analysis

Grama Jyothi needs a reliable electricity billing system that calculates monthly consumption and bills for households. It must avoid incorrect bills caused by floating-point precision errors, meter rollover, and inconsistent tariff values. The program calculates consumption from previous and current readings, applies the energy tariff slabs, and adds a fixed charge. The final bill must be accurate to the nearest paisa.

1.2 Input Data

Data item	Purpose	Suitable representation
Household number	Uniquely identifies each house.	unsigned int, or string if it contains letters
Previous reading	Meter reading at start of billing period.	meter_t (unsigned short)
Current reading	Latest meter reading.	meter_t (unsigned short)
Household category	Selects applicable tariff category.	enum or integer code
Tariff rates	Determine charge per unit.	Integer paise
Fixed charge	Amount charged regardless of consumption.	Integer paise

1.3 Output Data

Output	Purpose
Units consumed	Difference between readings, accounting for rollover.
Energy charge	Charge calculated using applicable tariff slabs.
Fixed charge	Fixed amount added to the energy charge.
Total bill	Energy charge plus fixed charge, displayed in rupees and paise.

1.4 Constants and Variables

Constants include the meter maximum value (65535), slab limits (100 and 200 units), energy rates (0, 225 and 450 paise), and fixed charge (5000 paise). Define these centrally so tariff revisions can be made in one place. Variables include household number, readings, consumption, energy charge, and total bill.

1.5 Negative, Fractional and Large Values

• Negative values: Valid consumption and bill amounts should not be negative. A lower current reading may indicate rollover or invalid data.
• Fractional values: Represent ■2.25 as 225 paise to avoid floating-point rounding errors.
• Large values: For district-wide totals, use a sufficiently wide integer type such as long long after checking the maximum possible value.

1.6 Rules for a Correct Bill

1. Consumption must account for meter rollover.
2. The first 100 units are free under the given tariff.
3. Units 101–200 cost ■2.25 per unit.
4. Units above 200 cost ■4.50 per unit.
5. The fixed charge is ■50 per household.
6. The bill must never be negative.
7. Total bill = energy charge + fixed charge.
8. Monetary calculations must preserve exact paise values.
9. Invalid readings and household records must be detected.

1.7 Assumptions

Each meter is a 16-bit counter with maximum reading 65535; at most one rollover occurs between consecutive readings; readings are cumulative; the stated tariff applies to every household unless a category specifies otherwise; the fixed charge is ■50 per billing period; money is displayed to two decimal places.

1.8 Alternative Concept: Store Readings as Strings

Readings such as “01530” could be stored as character arrays. This preserves leading zeros, but the text must be validated and converted before arithmetic. Parsing or comparing a string of d digits takes O(d) time; a null-terminated string needs approximately d+1 bytes. Numeric integer comparison and subtraction are O(1) for fixed-width values. Lexicographic comparison can be incorrect unless strings have a consistent format and length.